TOLGA ARSLAN

Çekmeköy, İstanbul | tolgaarslan.dev@gmail.com | +90 541 228 58 04
tolgaarslann.github.io | linkedin.com/in/tolga-arslan-dev | github.com/TolgaARSLANN

Summary

Software developer with a B.Sc. in Computer Engineering, building full-stack web applications with React, TypeScript, Node.js/Express and Python (FastAPI, Flask). Four portfolio projects, three live as public demos and backed by 700+ automated tests, span REST API design, secure authentication, SQL and NoSQL data modeling, CI, Docker deployment and ML-powered services. Internships in Agile teams; military service completed.

Technical Skills

Programming: JavaScript (ES6+), TypeScript, Python, SQL

Frontend: React, Vite, HTML5, CSS3, responsive and accessible UI

Backend: Node.js, Express.js, FastAPI, Flask, REST API design, JWT authentication

Databases: PostgreSQL, MongoDB, SQL Server, SQLAlchemy

Testing: Jest, Vitest, Playwright, pytest

DevOps & Tools: Git, GitHub Actions (CI), Docker, Vercel, Render

Data & ML: Pandas, NumPy, scikit-learn, LightGBM, OpenCV, PyTorch, Streamlit

Projects

Pawfect Care — Pet Health Management Platform *GitHub · Live Demo*

React, Vite, Node.js, Express.js, MongoDB, Jest, Vitest, GitHub Actions · capstone project, extended after graduation

- Built a full-stack platform for pet health records, vaccinations with booster alerts, weight trends, medications and vet appointments, plus a community blog and rule-based care recommendations
- Designed a 46-endpoint Express REST API with per-user data isolation, httpOnly SameSite=Strict JWT cookies, session revocation, brute-force rate limiting, CSP headers and input validation
- Delivered an accessible, responsive React UI with lazy-loaded routes, light/dark themes and keyboard support, plus account data export and deletion under a KVKK privacy notice
- Wrote ~300 automated tests (170+ Jest/Supertest, 120+ Vitest/React Testing Library) run with ESLint in GitHub Actions; deployed on Vercel and Render with MongoDB Atlas

Sudoku Solver — Photo-to-Solution Web App *GitHub · Live Demo*

React 19, TypeScript, Vite, Flask, OpenCV, PyTorch, ONNX, Docker, pytest, Playwright

- Engineered a computer-vision pipeline that flattens the grid in a phone photo, reads digits with k-NN, solves the puzzle with a bitmask backtracking solver and writes the answer back onto the photo
- Solved 37 of 40 unseen newspaper photos with zero wrong solutions; photos that cannot be read reliably are flagged to the user instead of guessed
- Trained a convolutional neural network in PyTorch on augmented MNIST and served it via ONNX to check puzzles filled in by pen, flagging ~90% of wrong digits with 3 false alarms in 2,073
- Shipped a React/TypeScript frontend and Flask API in a non-root Docker image on Render (~275 MB peak under a 512 MB limit); 111 pytest tests plus Playwright end-to-end tests

HavaUyarı — Air Quality Early Warning System *GitHub*

Python, FastAPI, PostgreSQL, SQLAlchemy, Alembic, Docker, Streamlit, LightGBM, pytest, GitHub Actions

- Developed an end-to-end system that forecasts PM2.5 24 hours ahead for 8 stations in 5 Turkish cities and warns before unhealthy air, using Ministry SİM measurements and three Open-Meteo APIs
- Cut MAE from 12.9 to 6.8 µg/m³ in a 12-month walk-forward backtest over 62.6K station-hours: 47.7% below the raw CAMS forecast and 25.4% below the best simple baseline
- Doubled alert precision from 0.36 to 0.73 while raising recall from 0.25 to 0.66; the Streamlit dashboard shows 80% prediction intervals and SHAP-based explanations
- Served forecasts via FastAPI with an hourly cache, PostgreSQL storage and Alembic migrations on Docker Compose; 152 tests, including data-leakage and PostgreSQL integration tests, run in CI

Card Fraud Detection — Real-Time Transaction Scoring *GitHub · Live Demo*

Python, FastAPI, LightGBM, XGBoost, SHAP, Streamlit, Docker, pytest, GitHub Actions

- Trained a fraud model on 1.85M synthetic card transactions (Kaggle Sparkov) with point-in-time features and a pre-registered, time-based test; LightGBM reached 0.959 PR-AUC vs. 0.534 for logistic regression
- Designed an expected-cost alert rule: over 6 test months, 2,522 alerts at 74% precision caught 87% of fraud cases and 98% of fraud value, cutting total cost from $1.13M to $45.5K
- Built a FastAPI scoring service returning a decision and the top 3 SHAP reasons per transaction, with feature parity between training and serving verified on 2,000 transactions
- Traced unbounded memory growth in the Docker demo to glibc malloc arenas and fixed it, holding memory at ~670 MB with 3 sessions instead of 936 MB and rising; 176 tests in CI

Experience

ArVis Technology – DeepTech/AI Intern · *Teknopark İstanbul* *July 2024 – Aug 2024*

AI R&D internship program run in collaboration with Roket Technology and Cubeforward

- Prepared and preprocessed training data for the CNN model of "Neurovisionist", an AI-powered early Alzheimer's diagnosis project, within a 16-person R&D team
- Developed cognitive-exercise mini apps targeting memory functions in Agile sprints

VBT Yazılım A.Ş. – Software Intern *July 2023 – Aug 2023*

- Completed a 6-week training program of 50+ sessions led by VBT engineers, focused on SQL Server: joins, views, user-defined functions, stored procedures, indexing and backup/restore
- Built practice applications with C#, .NET Core and SQL Server, using Git for version control
- Attended sessions on web (HTML, CSS, JavaScript, Angular), Flutter, Redis, RabbitMQ and ITSM/ITIL

Education

Bandırma Onyedi Eylül University *Sept 2020 – Feb 2025*

Bachelor of Science in Computer Engineering (English-taught program) *Bandırma, Türkiye*

- Coursework: Data Structures & Algorithms, Object-Oriented Programming, Databases, Software Engineering

Certifications & Training

Certificate of Achievement, DeepTech/AI Internship Program · *ArVis Technology, 2024*

Data & AI Class · *İş Bankası ProSchool, in partnership with Toptalent.co*

SQL Server Database Programming · *Udemy*

Languages

Turkish: Native · **English:** B2 (Upper-Intermediate), English-taught B.Sc. program